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CC 207B Unit 5 Assignment

Name: Caryl Ann Calibjo Date: 10/13/26

Year and Section: BSIT 3B

Note: Upload your Flutter project to your assigned folder here -> https://github.com/WVSU-BSIT-Batch-2028/CC-207B_26-27_Activities. Read the instructions on submitting, including the file name/folder conventions in the GitHub repository.

I. CICT Org Fair App Development. The CICT is holding an Org Fair. Students browse the student organizations on their phone, open one to read about it, and sign up with a short form. Most of this app is already written for you. Your job is to **finish 14 small pieces**, **explain how the rest works**, and **run experiments** that show what each Flutter idea from 5.1 and 5.2 really does. You will write about 40 lines of code. **(35 points)**

Get the initial Flutter project source code from: https://github.com/WVSU-BSIT-Batch-2028/CC-207B_26-27_Samples/tree/main/Unit%205/assignment_org_fair

Here is the table specifying the code changes you need to perform:

File	What it does	What you do
`lib/main.dart`	Starts the app and sets the theme	Read (E1)
`lib/data/orgs.dart`	The `Org` class and a pretend web service, `OrgRepository`, that waits about 2 seconds and returns orgs, nothing, or an error	Read only. Do not change
`lib/models/registration.dart`	The `Committee` enum and the `Registration` class the form returns	Read
`lib/widgets/state_message.dart`	One layout for the loading, error, empty and no-results messages	Read
`lib/widgets/org_card.dart`	One org in the list	Read (E6)
`lib/widgets/org_banner.dart`	The banner photo, with loading and error handling	Read (E8)
`lib/screens/org_list_screen.dart`	Loads the orgs, search, list or grid, swipe and long-press	TODO 1 to 7; E2 to E5, E7

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File	What it does	What you do
`lib/screens/org_detail_screen.dart`	One org with its banner and a Join button	TODO 8; E9
`lib/screens/join_form_screen.dart`	The sign-up form with 8 kinds of input	TODO 9 to 14; E10, E11
`test/org_fair_test.dart`	7 automatic tests. Each one names the TODOs it checks	Run to check your work

Note: Run the command “flutter test” at any time. All tests should pass before submitting the assignment.

Each TODO is a comment in the code with a placeholder line under it. Replace the placeholder with your code. Most need 1 to 6 lines. In `org_list_screen.dart` the loading state is already written; use it as a pattern for TODO 4 and 5.

#	File	What to do	Slide topic
1	org_list_screen	Start loading the orgs once in `initState()`	StatefulWidget Lifecycle
2	org_list_screen	`_reload()`: fetch again inside `setState()`	What Does setState() Do?
3	org_list_screen	`_toggleFavorite()`: add or remove the id inside `setState()`	setState; 5.1 Set
4	org_list_screen	Error state with a **Try again** button	The Four States of Content
5	org_list_screen	Empty state with a **Refresh** button	The Four States of Content
6	org_list_screen	Search box `onChanged` saves the text with `setState()`	Getting Values from Users
7	org_list_screen	Wrap each card in a `Dismissible`	Dismissibles
8	org_detail_screen	Join button: disabled when already a member	Affordance and Feedback
9	join_form_screen	`validate()` then `save()` the form	Handling forms with validation
10	join_form_screen	Send the registration back with `Navigator.pop`	Navigation widgets
11	join_form_screen	Email validator with helpful messages	When Should Errors Appear?
12	join_form_screen	Student ID: digits-only formatter and validator	Input Validations (formatters)
13	join_form_screen	Slider, Switch and Checkbox `onChanged` with `setState()`	Inputs Follow One Loop
14	join_form_screen	Submit stays disabled until the checkbox is ticked	Affordance and Feedback

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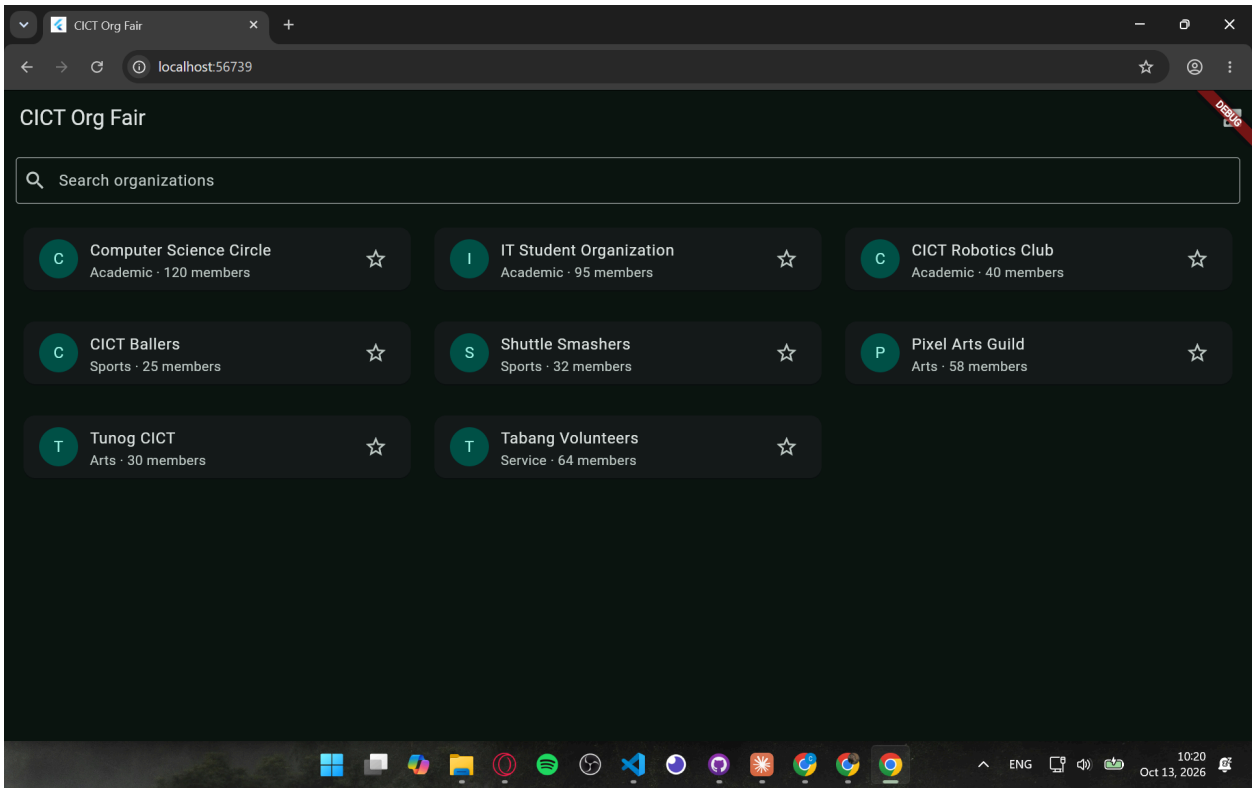
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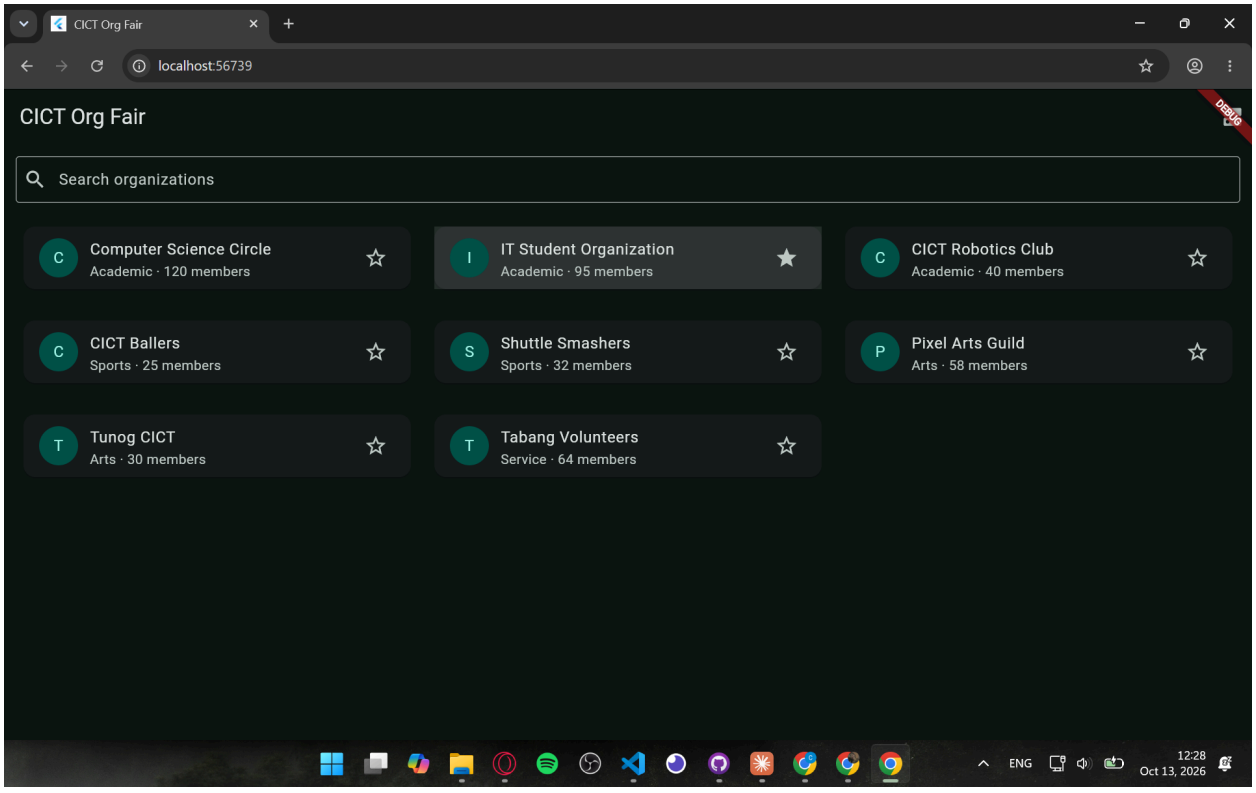
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Evidence (Provide screenshots of the application to show that it is working):





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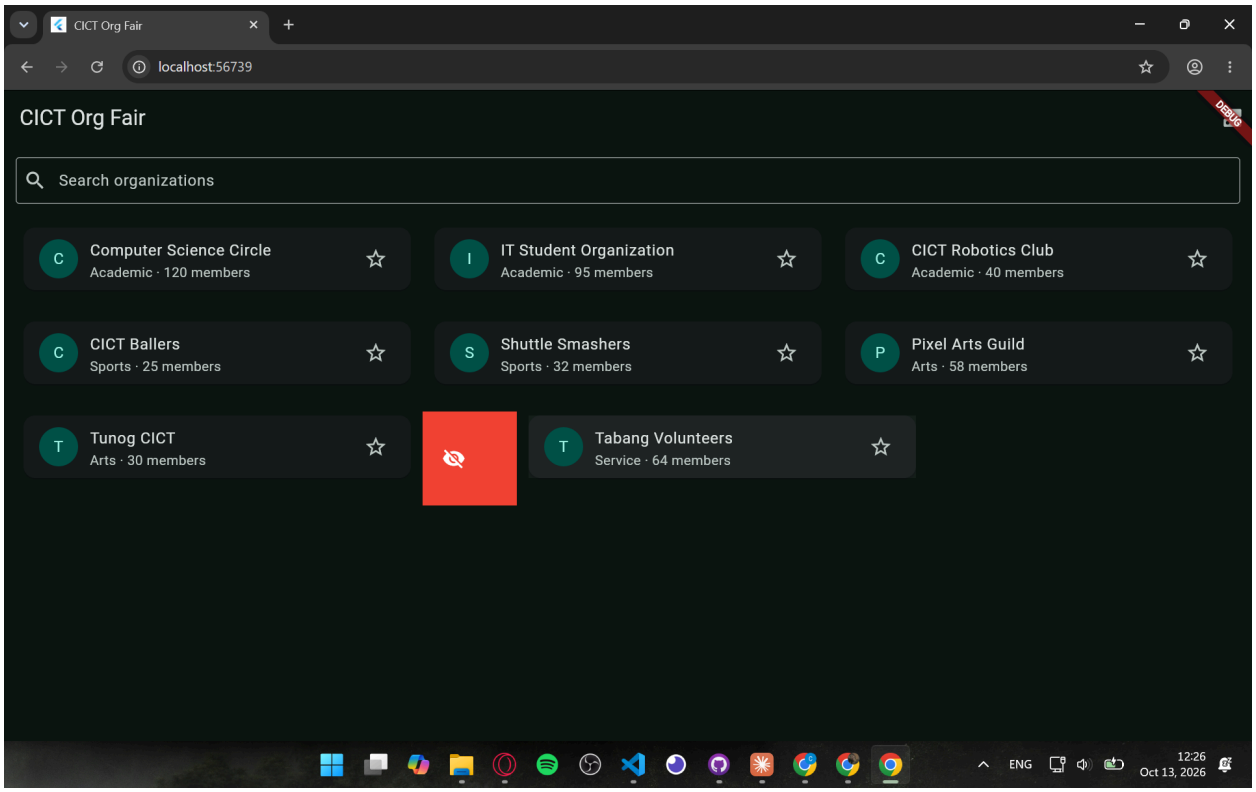
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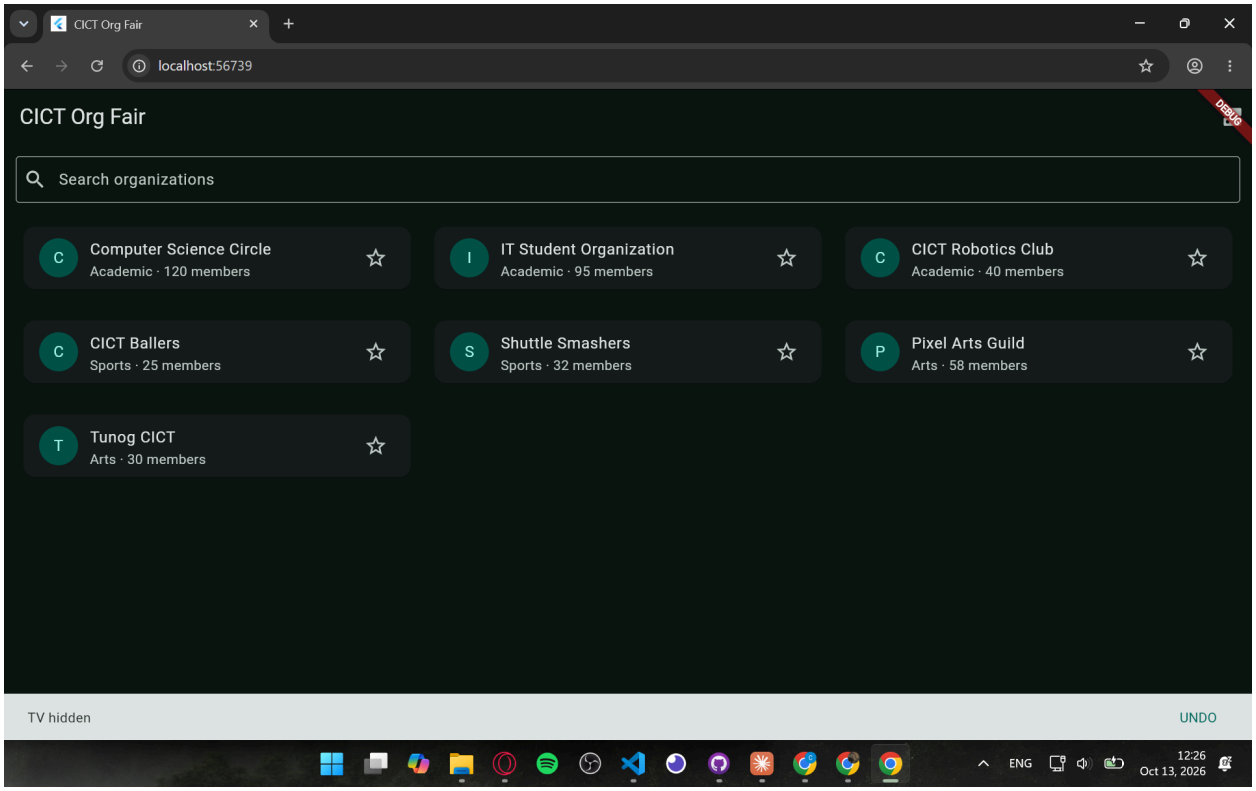
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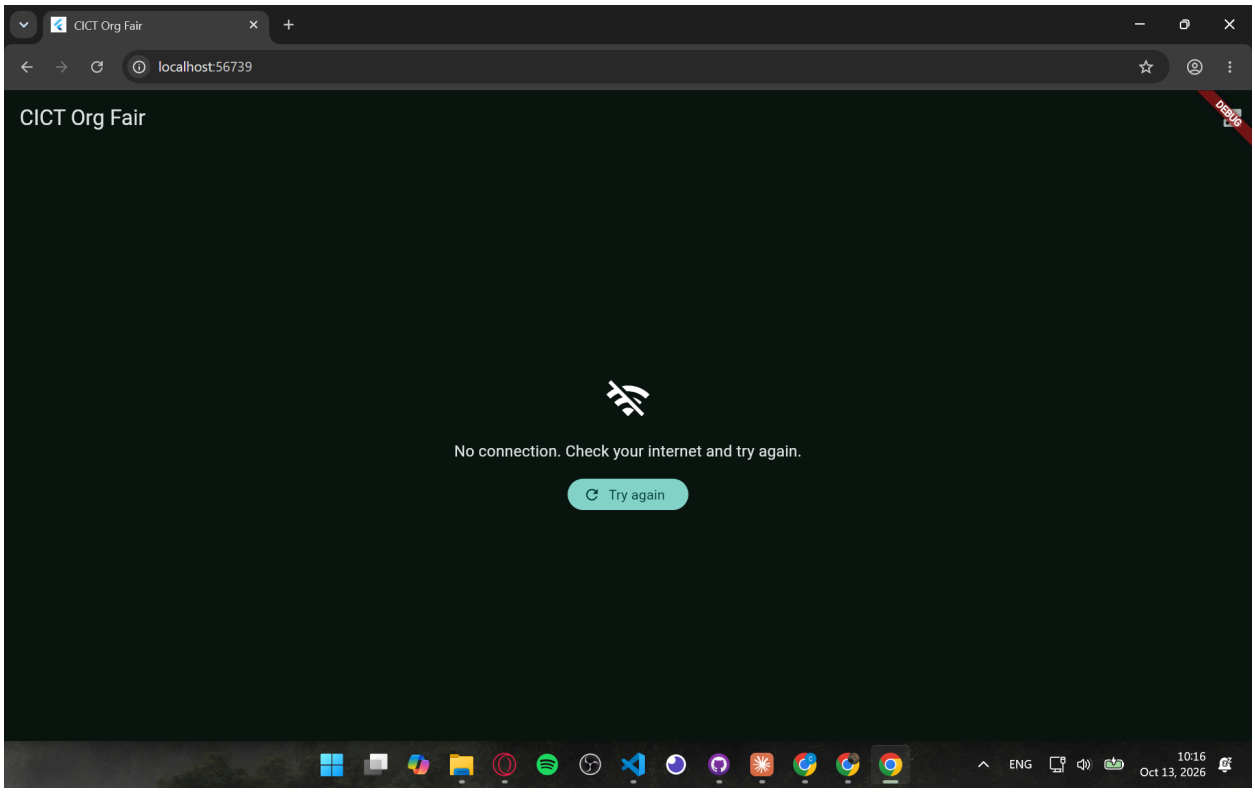
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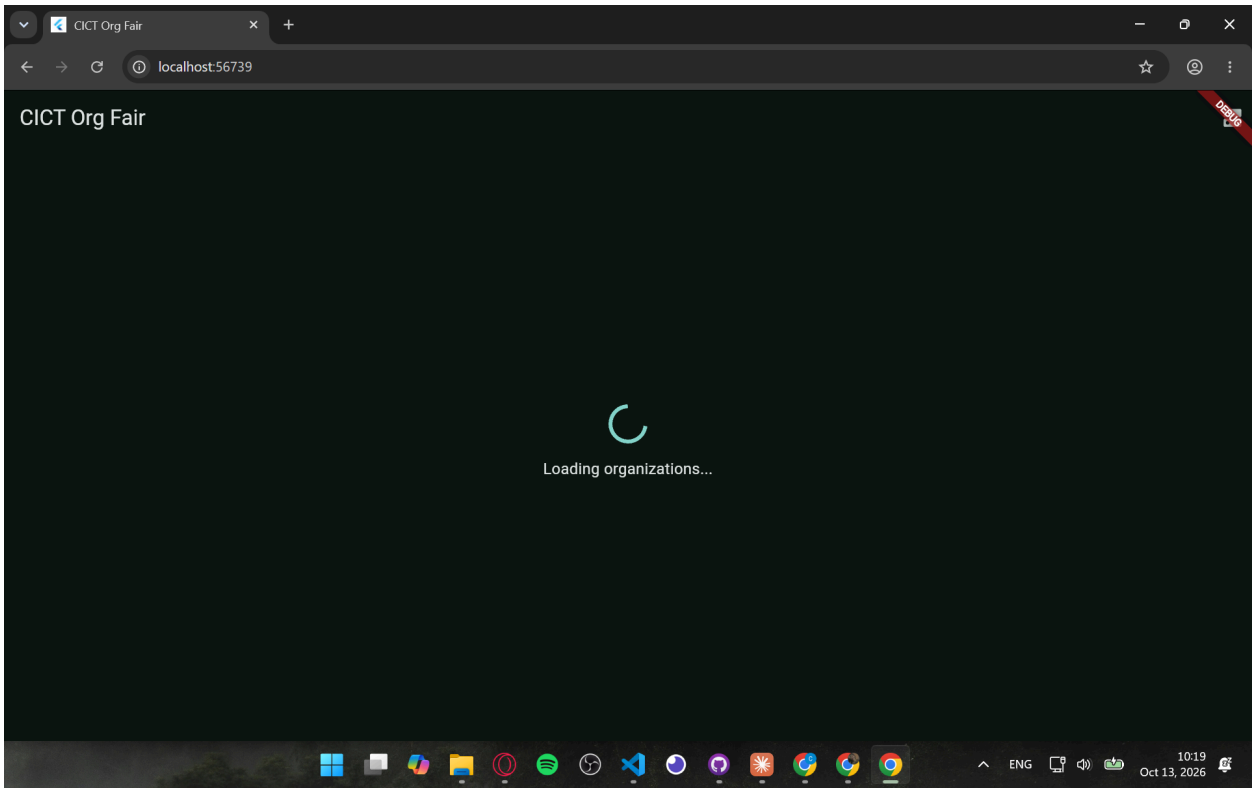
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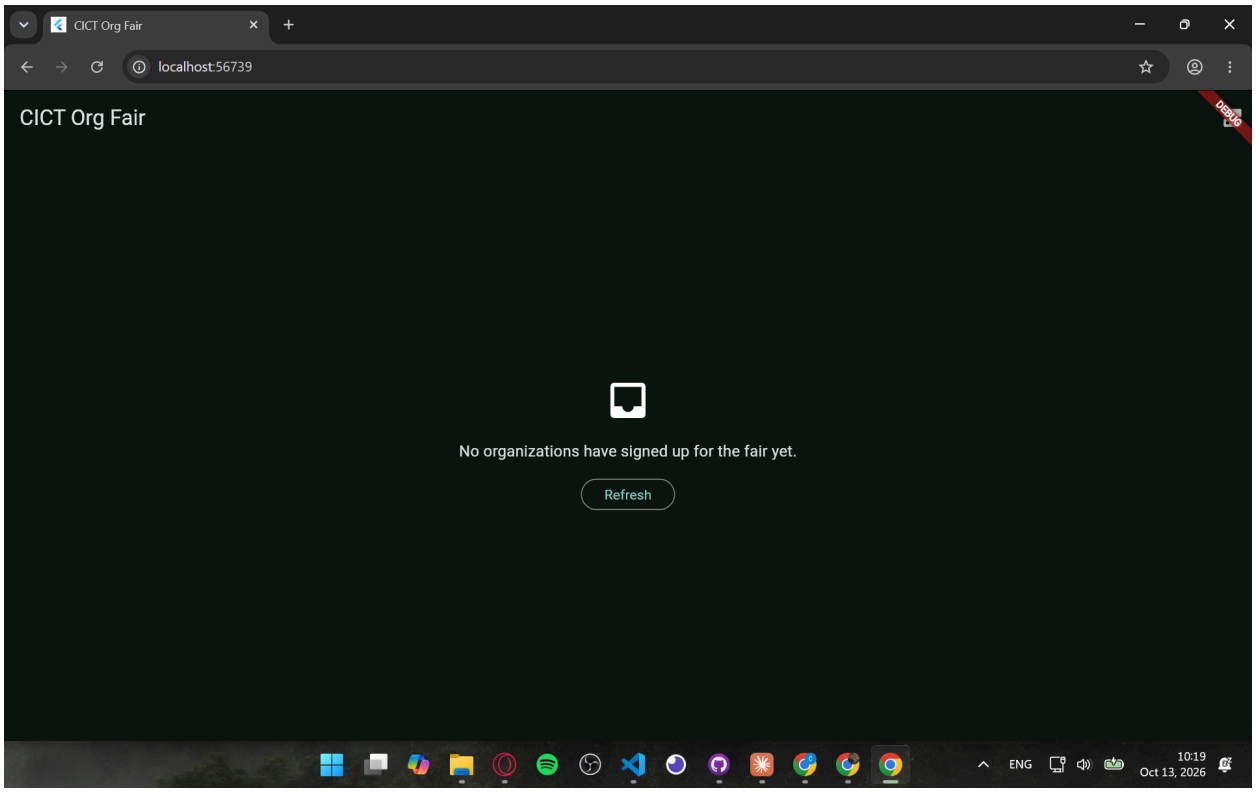
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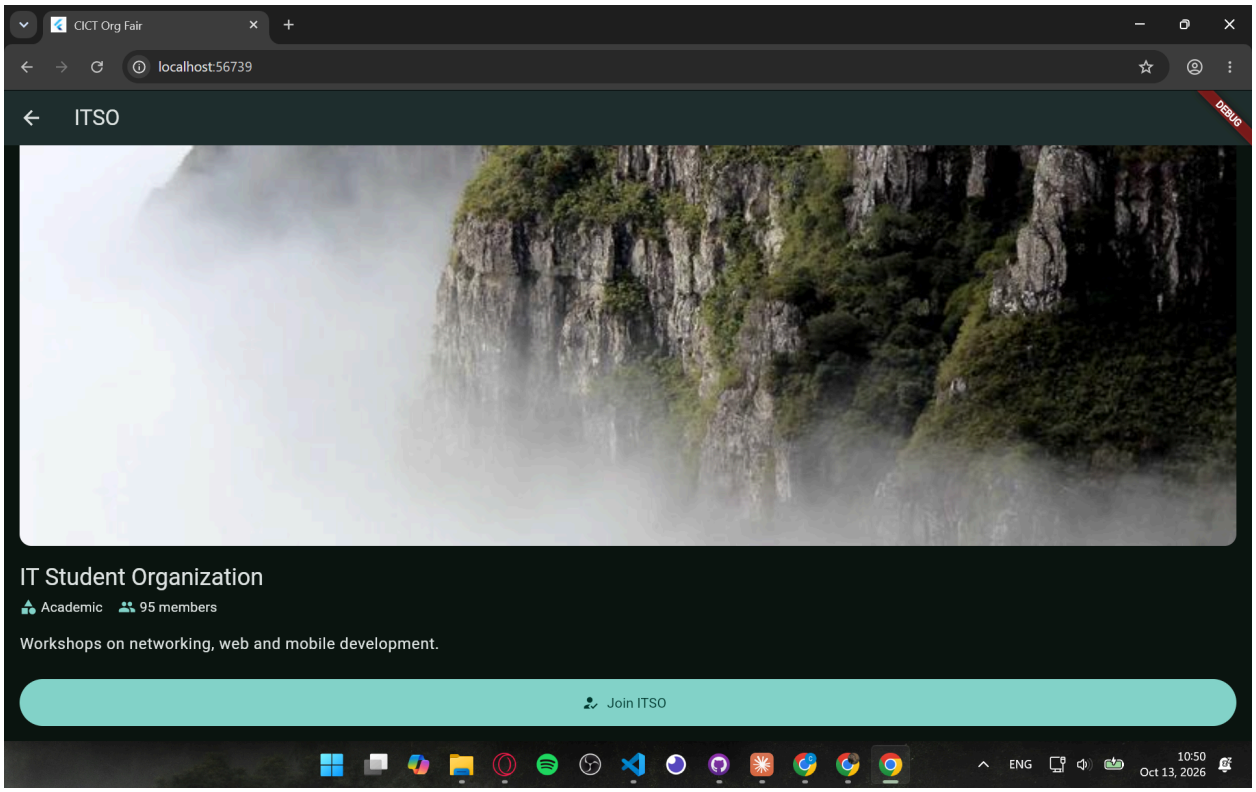
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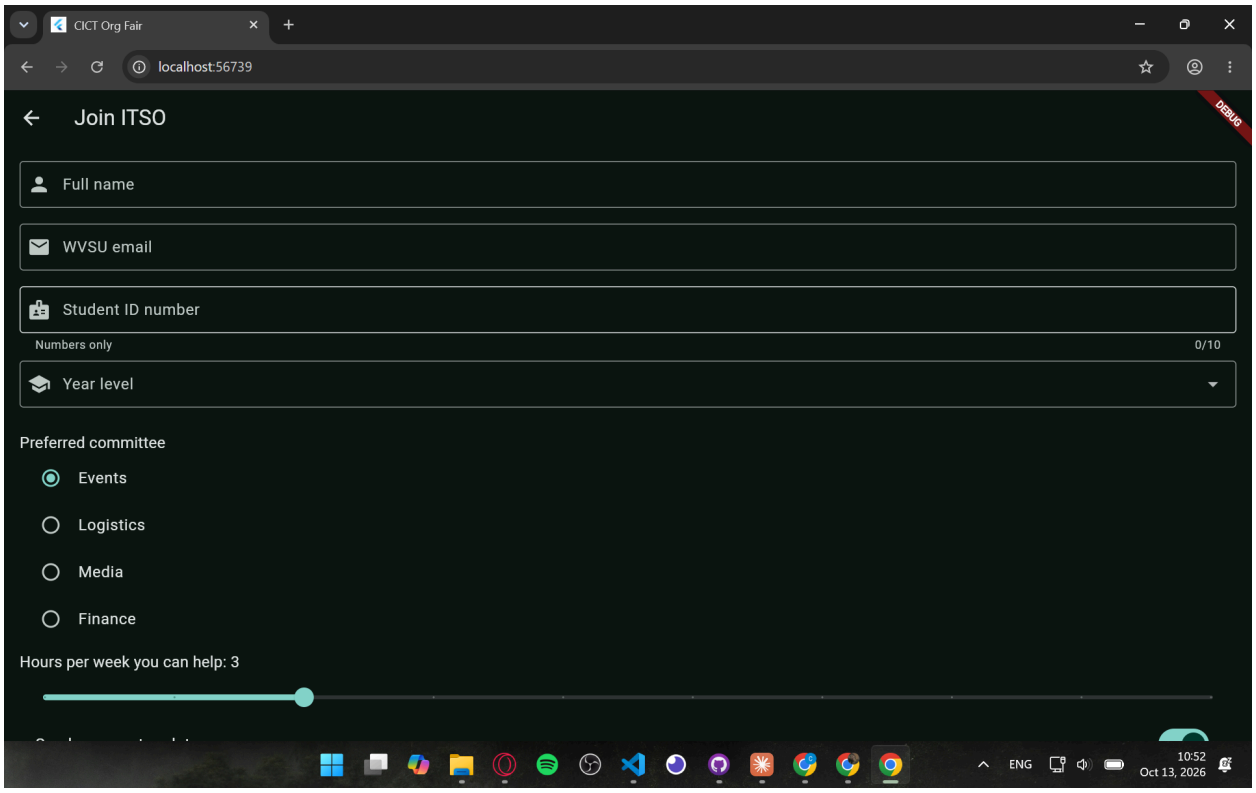
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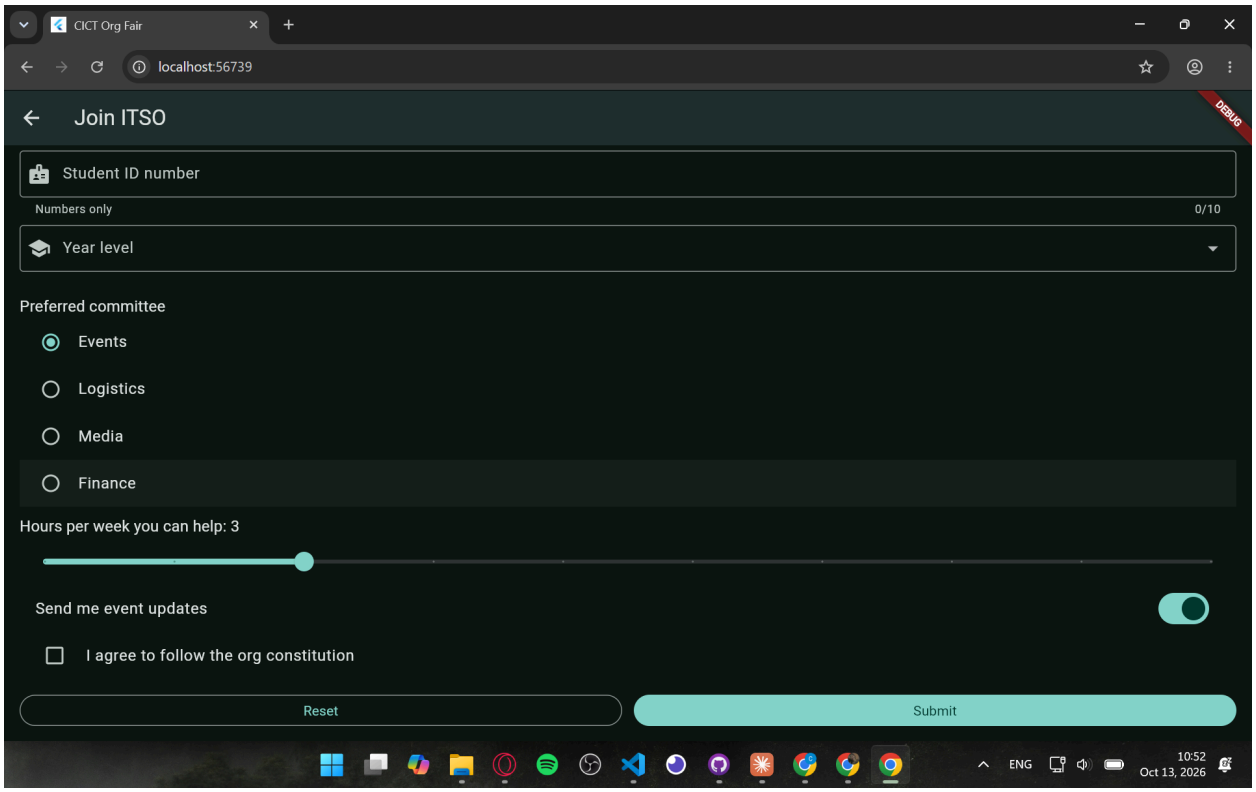
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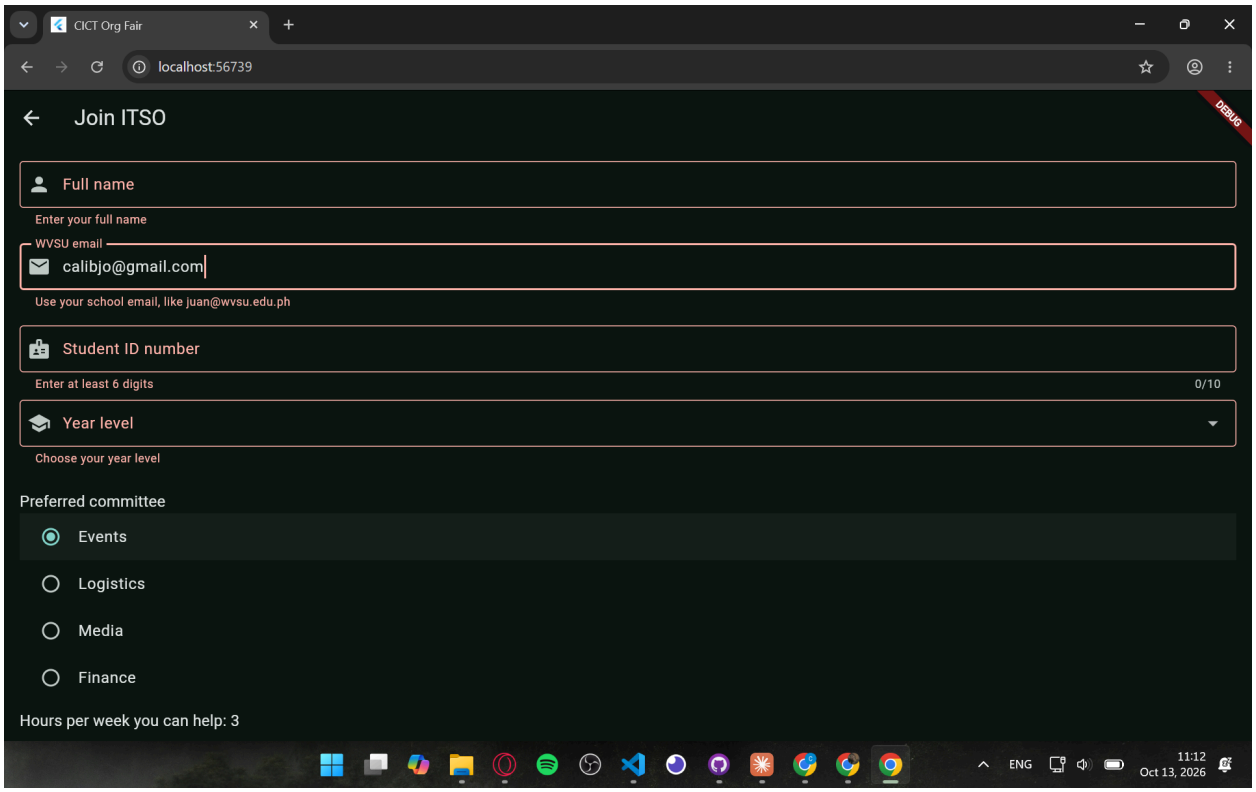
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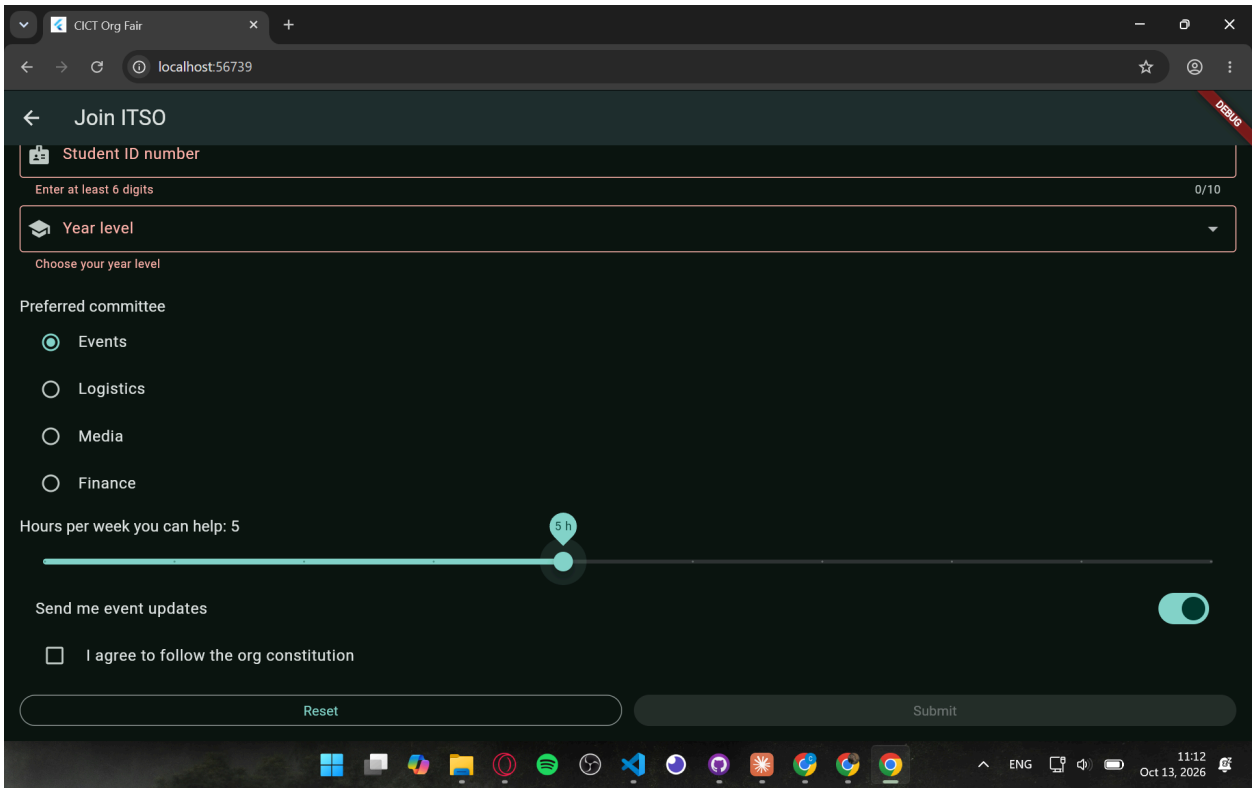
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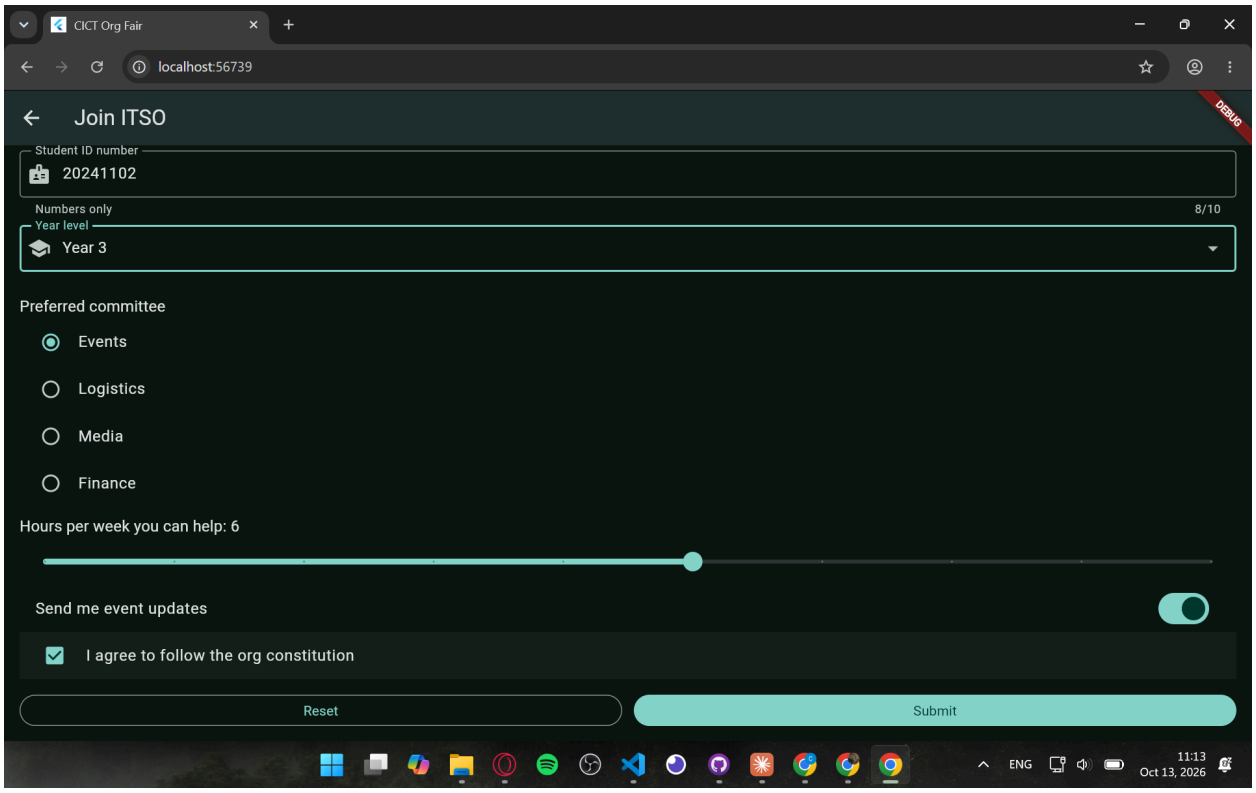
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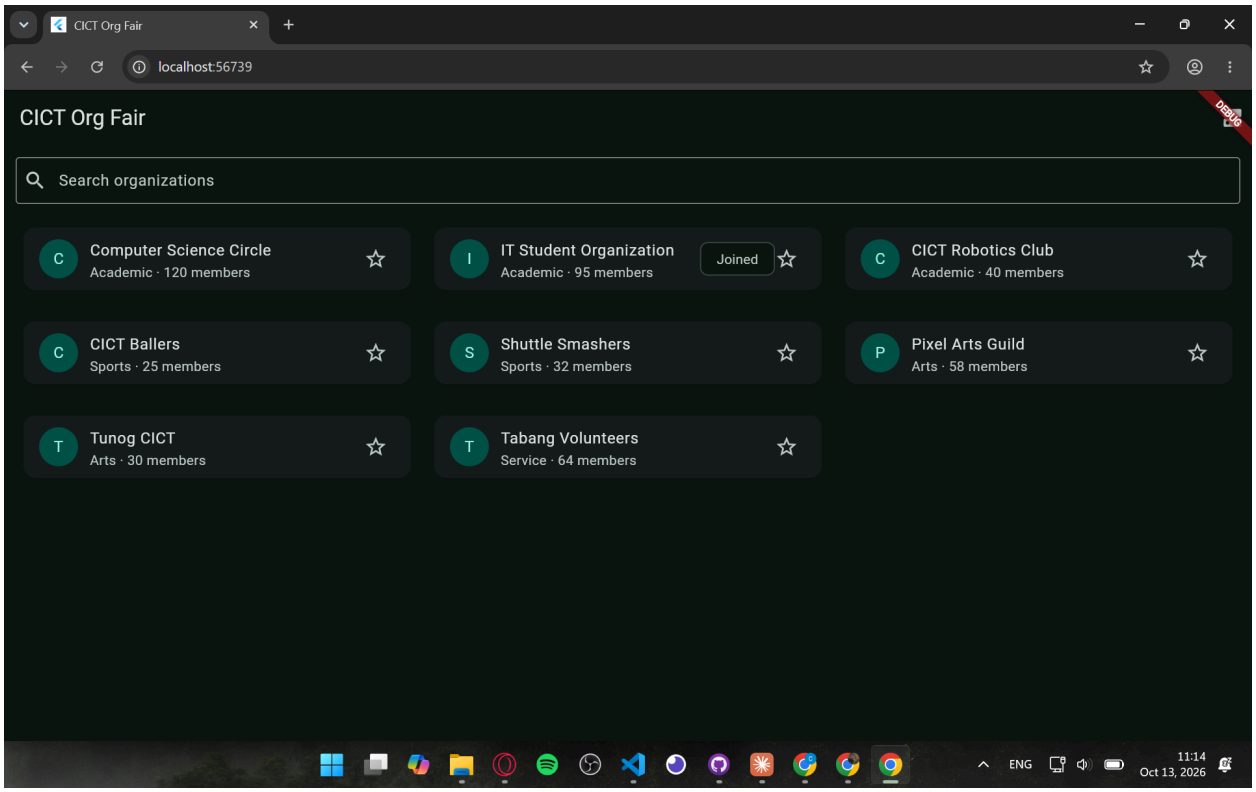
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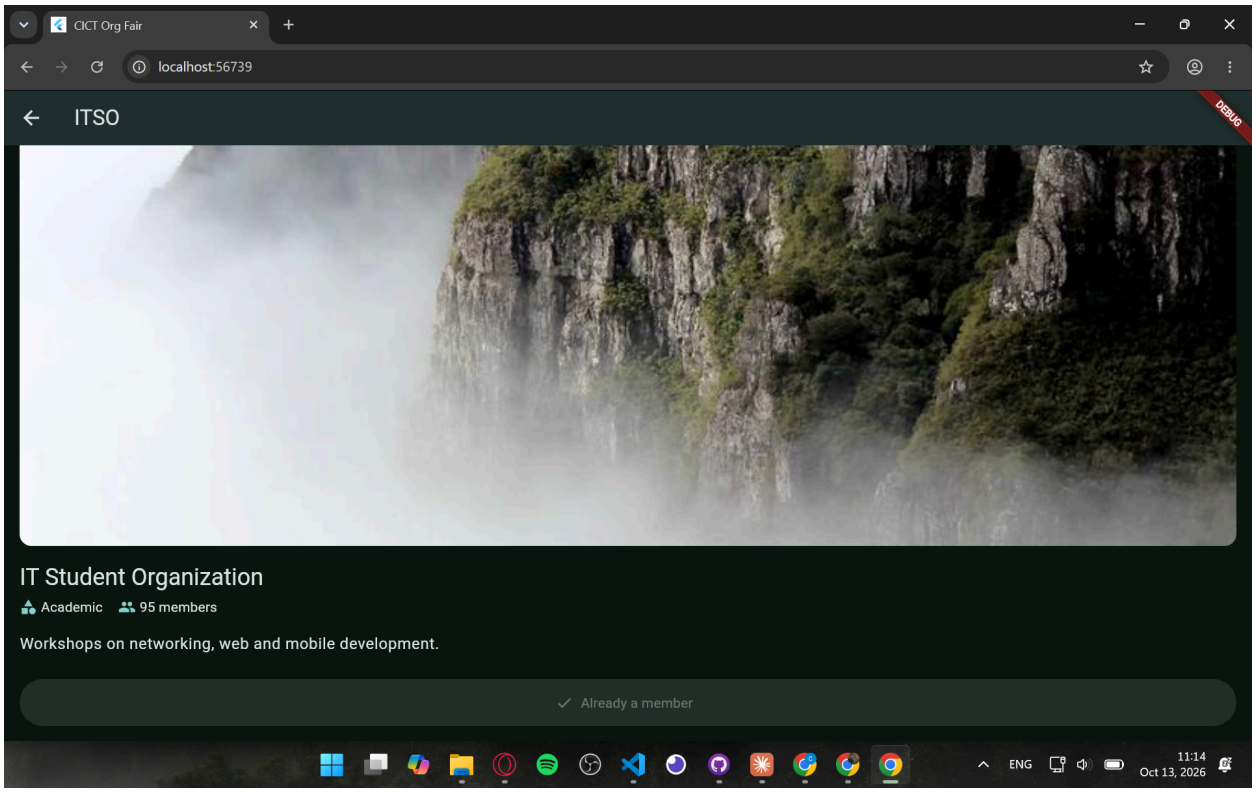
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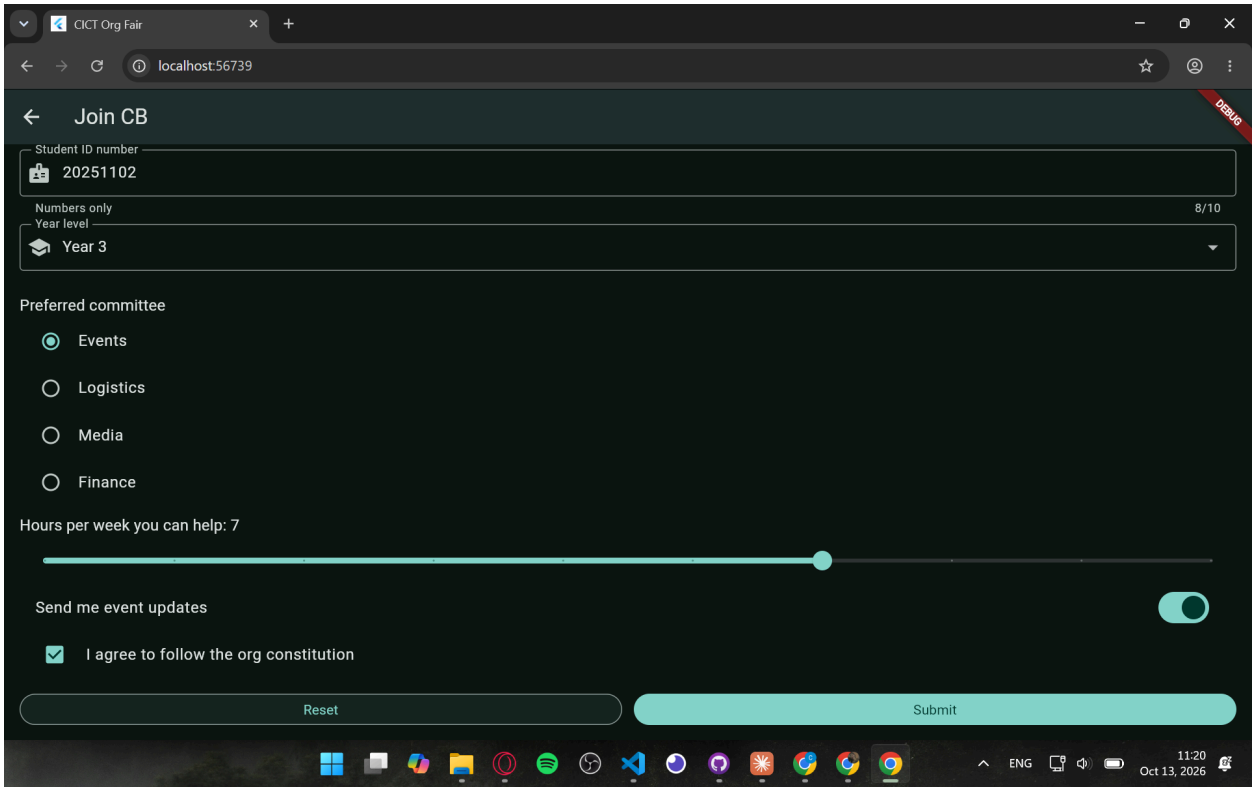
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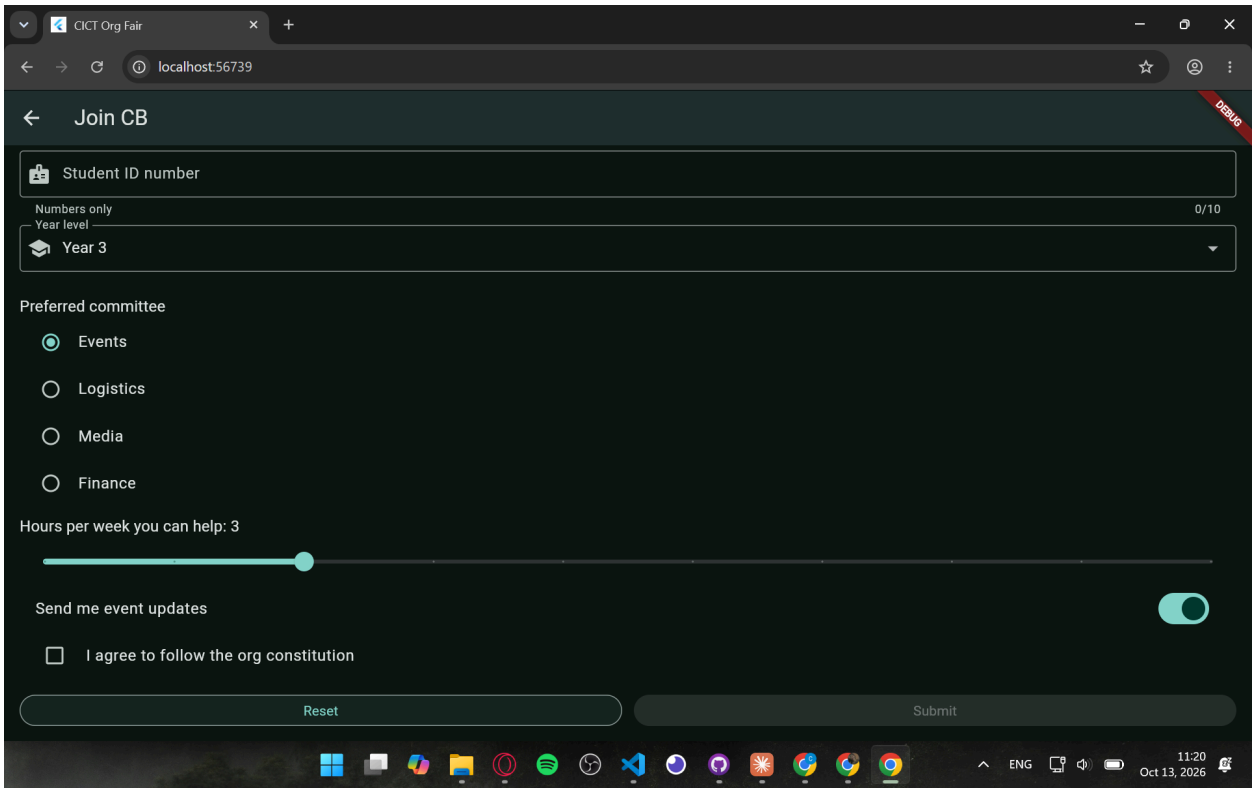
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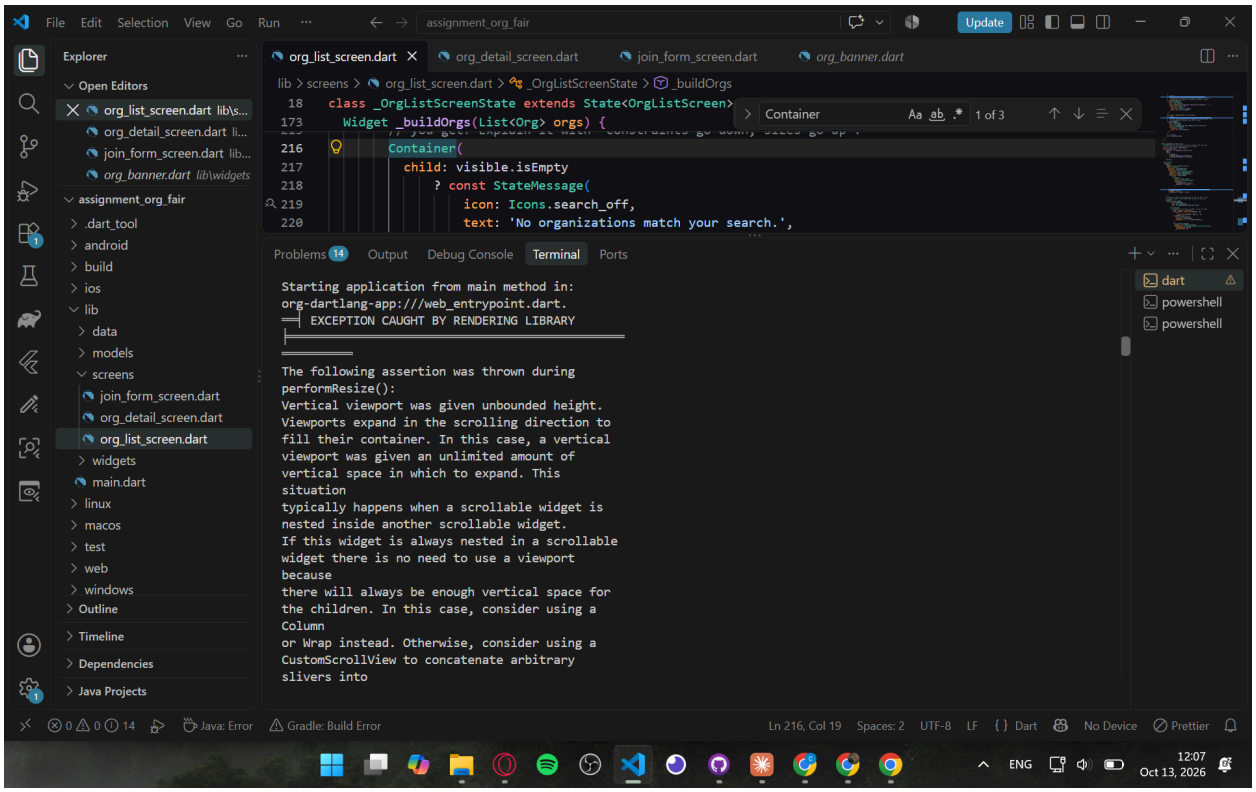
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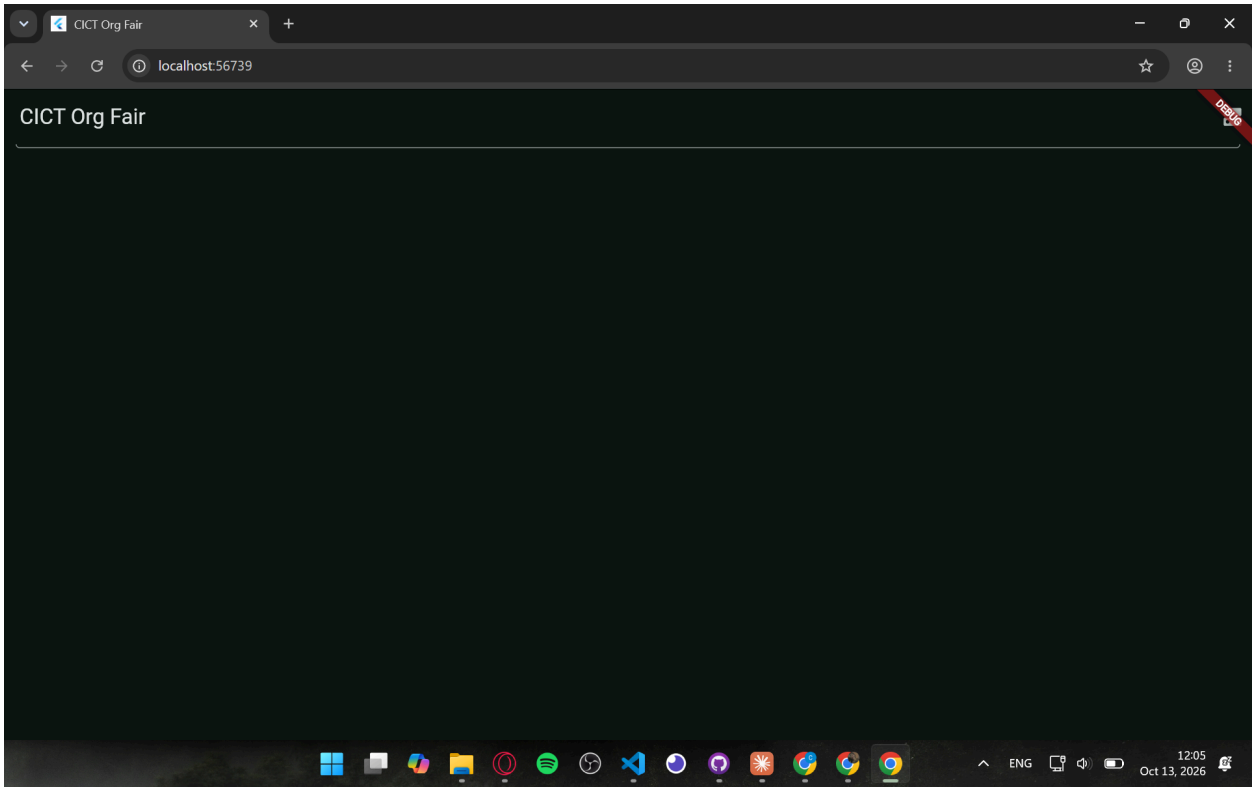
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Part II. E5





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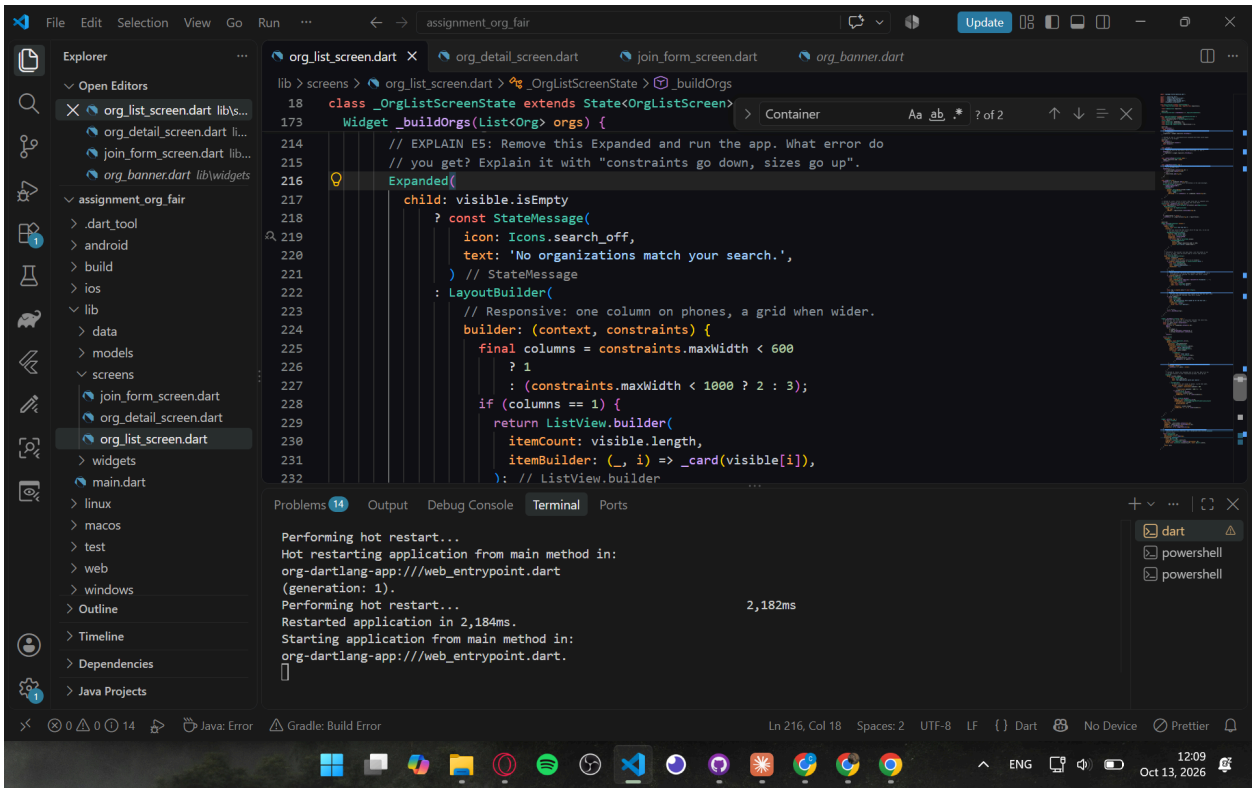
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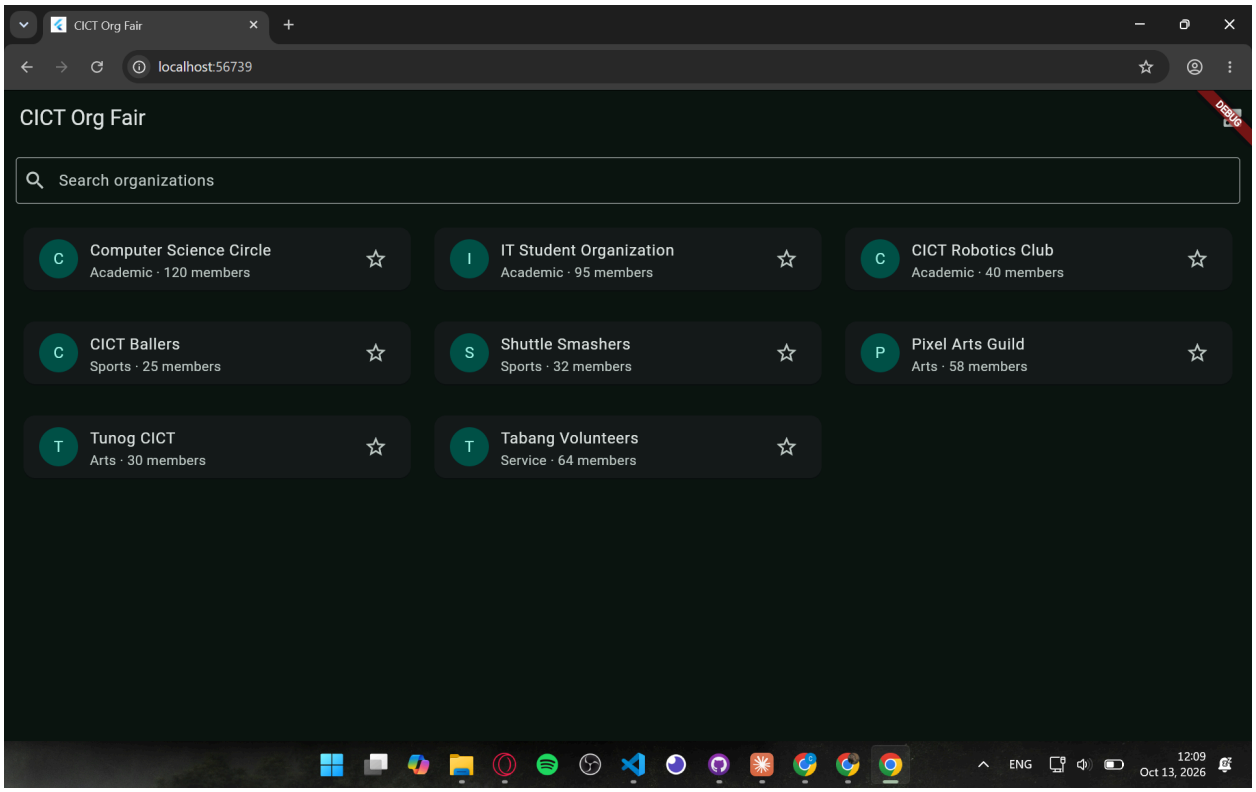
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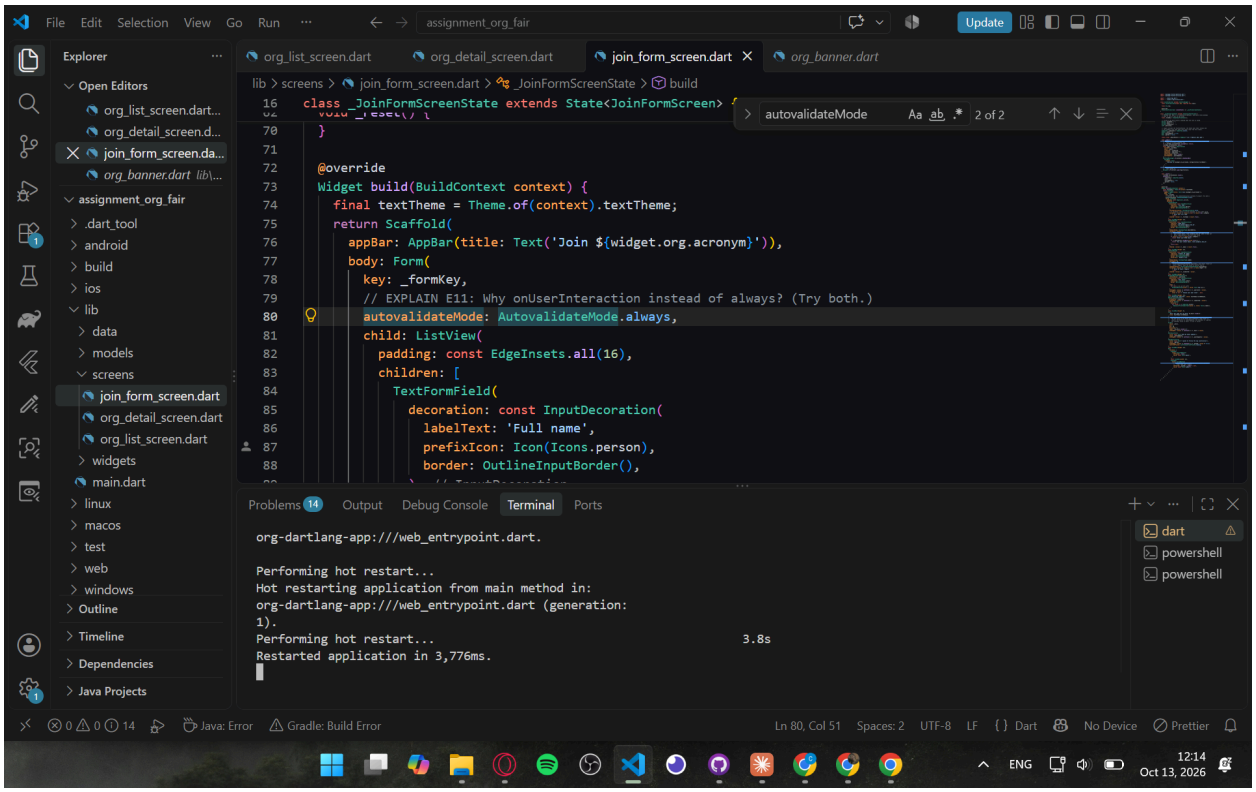
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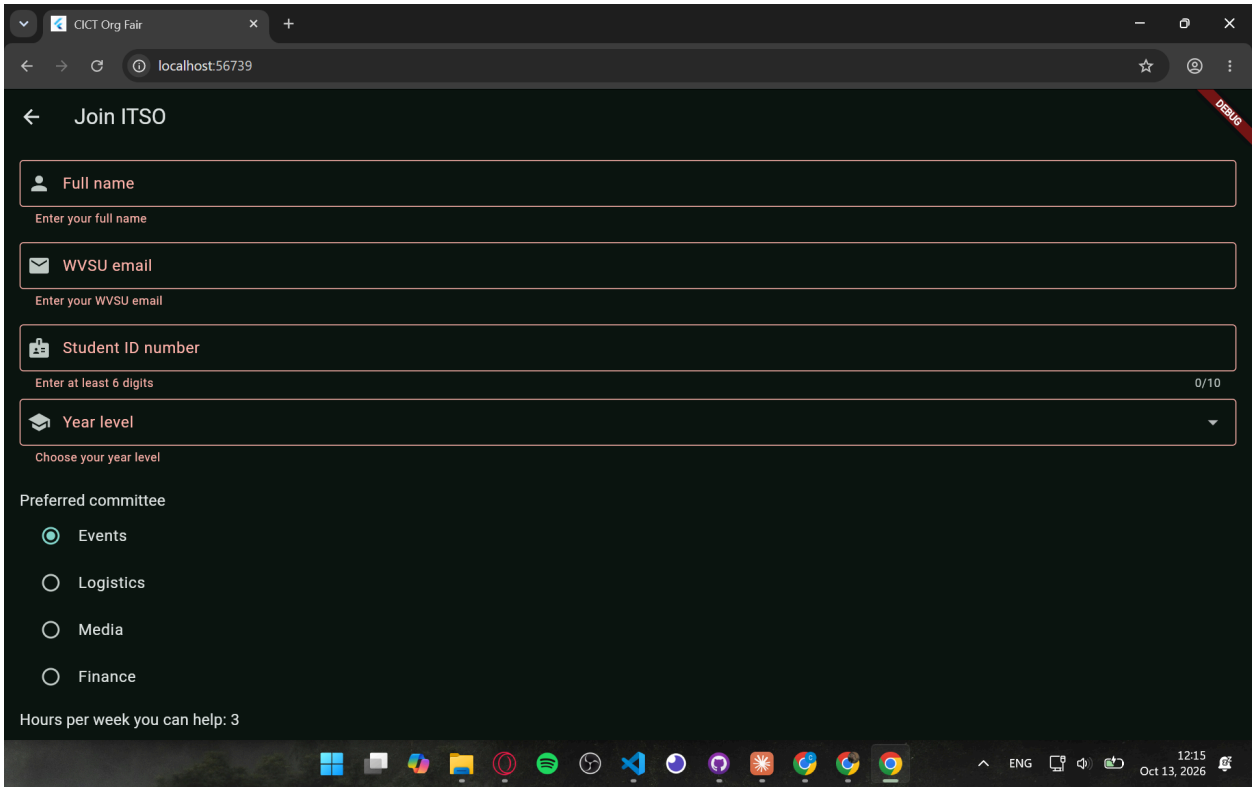
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Part II. E11





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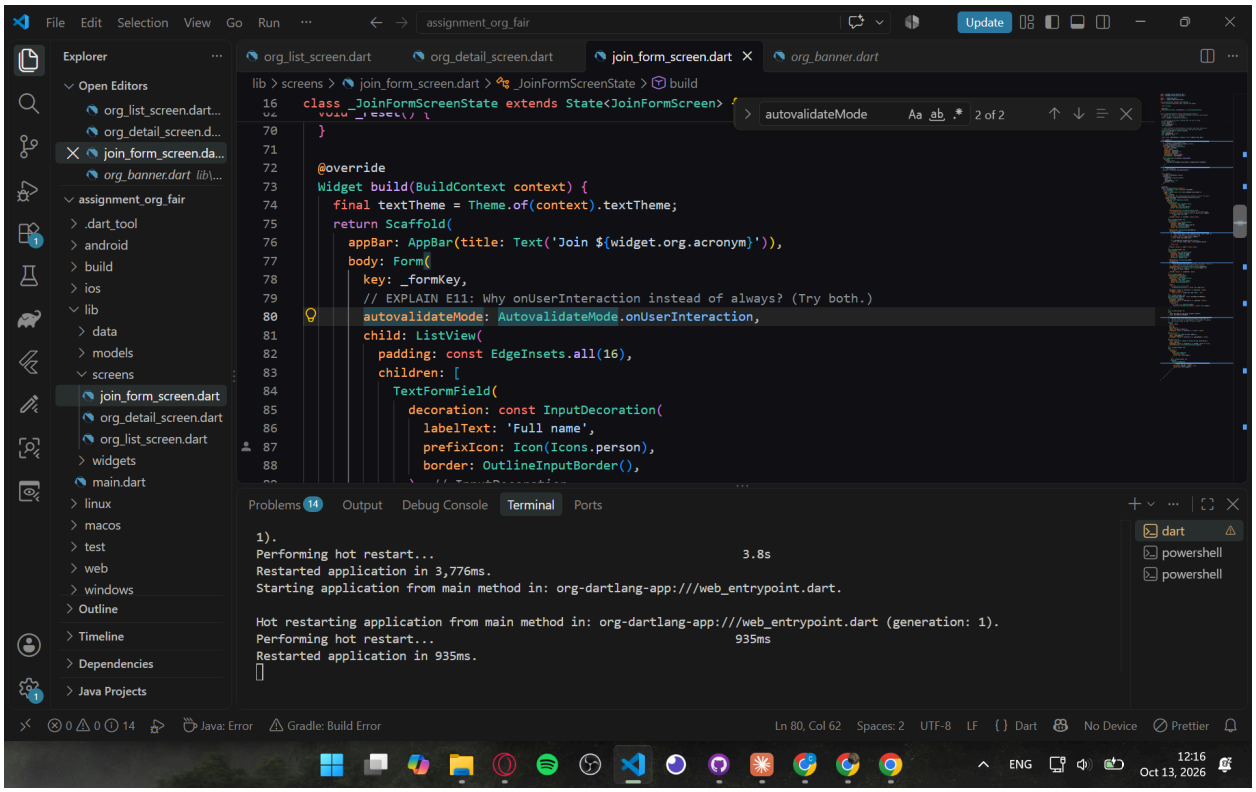
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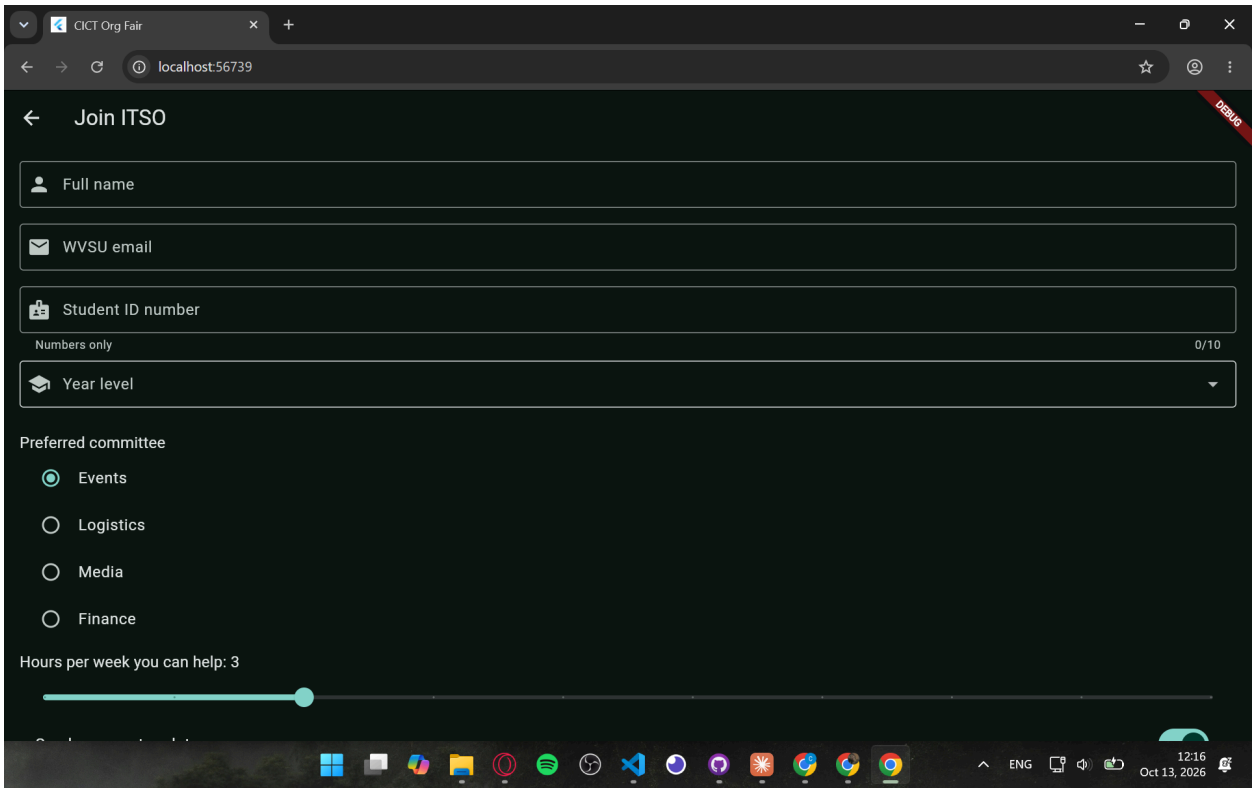
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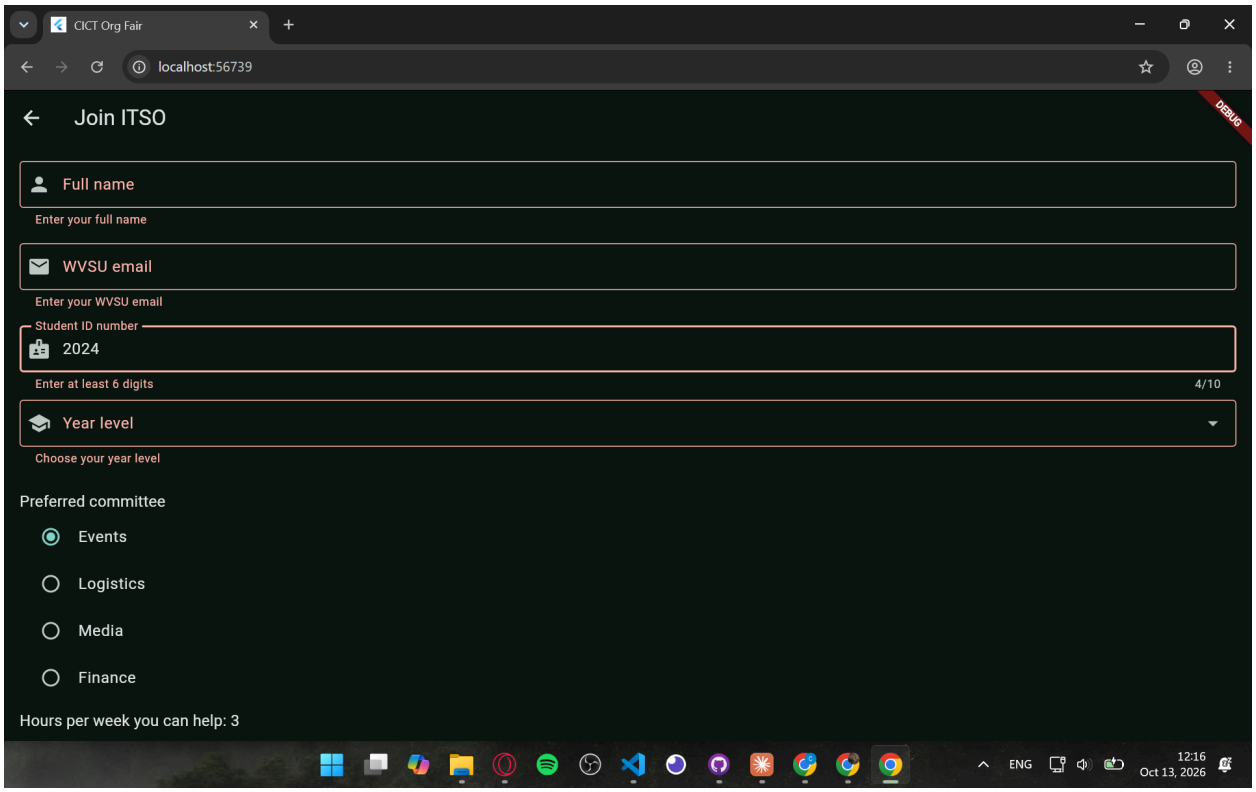
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II. CICT Org Fair Code Analysis. The code has 11 comments that start with `EXPLAIN E1` to `EXPLAIN E11`. Put your answers in the available column for each row. (2 points each)

#	Where	The question is about	Answer
E1	main.dart	Why OrgFairApp is stateless but OrgListScreen is stateful?	OrgFairApp never changes. It receives the repository once and builds the same app and theme every time. It owns no data that changes, so it is Stateless. OrgListScreen holds data where its content changes while visible: loading, search text, favorites, and hidden orgs. The screen itself causes these changes through typing, long press, swipe, and Try again. It needs a State object, so it is Stateful.
E2	org_list_screen: dispose()	Why controllers must be disposed?	**dispose()** runs once at the end of the screen's life. A controller holds resources and listeners. Flutter does not free them on its own. Without **dispose()**, each closed screen leaves a controller in memory. The leftovers pile up and slow the app. Disposing frees the controller when the screen leaves.
E3	org_list_screen: FutureBuilder	The order of states when the user presses Try again?	1. Done with an error. The error message and the Try again button show. 2. The user presses the button. **_reload()** calls **setState()** with a new Future. 3. Waiting. The loading spinner shows for about 2 seconds. 4. Done with an error again, because the server mode is still "error". If the mode is normal, the state is done with data and the list shows.
E4	org_list_screen: _buildOrgs	How the list updates while the user types?	No loop watches the search box. The TextField calls onChanged on every keystroke. onChanged saves the text in **_query** with setState(). **setState()** marks the screen dirty, so **build()** runs again. build() calls **_buildOrgs()**, which filters with the new _query. The list updates on every keystroke.

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#	Where	The question is about	Answer
E5	org_list_screen: Expanded	Constraints go down, sizes go up?	Without Expanded, the Column gives the list no height limit. A ListView asks for all the height it can get, so it asks for infinite height. Flutter then throws "Vertical viewport was given unbounded height". Constraints go down: the parent tells the child how much space it may use. Sizes go up: the child picks a size inside those limits. Expanded gives the list the leftover height of the Column. The list gets a limit, picks its size, and scrolls inside it.
E6	org_card.dart	Where the favorite star's state lives?	The star changes while visible (Q1 yes), but the card does not cause the change (Q2 no). So OrgCard stays stateless. The state lives in _favoriteIds, a Set in the State of OrgListScreen. OrgCard receives isFavorite as a plain bool and draws a filled or empty star. A long press calls onLongPress, which runs _toggleFavorite(). That calls setState(), and the parent rebuilds the card with a new isFavorite value. The widget only shows the value. The State owns it.
E7	org_list_screen: _openDetail	How data comes back from another screen?	**push()** returns a Future. The Future completes with the value passed to **Navigator.pop()** when the pushed screen closes. It completes with null if the user presses back. Here the value is a Registration. The line **Navigator.of(context).pop(registration);** in _submit() of the form screen sends it back. The detail screen passes it on with its own pop. Then await in **_openDetail** receives it, and setState() saves it in _registrations.

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#	Where	The question is about	Answer
E8	org_banner.dart	Null safety and the loading and error builders?	No URL: if (url == null) return _placeholder(context, Icons.groups); handles it. The user sees a colored box with a groups icon. Still downloading: loadingBuilder handles it. The user sees a spinner until progress is null. Failed: errorBuilder handles it. The user sees a colored box with a broken image icon. bannerUrl is String?, so it can be null. The code copies it to a local variable named url and checks it for null. After the check, Dart knows url is a String. This is type promotion. The compiler forces the code to handle the null case before Image.network runs.
E9	org_detail_screen: _join	Why check context.mounted after await?	A BuildContext is the location of a widget in the tree. An await pauses the function while the form screen is open. The user can close the screen during that pause, and the widget leaves the tree. The old context then points to nothing, and using it can crash the app. The context.mounted check confirms the widget is still on screen before Navigator.pop() runs.
E10	join_form_screen: _formKey	What a GlobalKey<FormState> gives you?	The key gives code outside the Form a reference to the Form's state. It allows three calls: - **validate()**: runs every field's validator. - **save()**: runs every field's onSaved. - **reset()**: sets every field back to its initial value.
E11	join_form_screen: autovalidateMode	When errors should appear?	With **AutovalidateMode.always**, errors show before the user types anything. The form feels like scolding. With onUserInteraction, a field shows its error only after the user edits it. Errors then appear when they help. This is usually the best choice, so the code uses it.

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Determine what type of widgets and whether they are either stateless or stateful. (2 points each)

Feature	Widget(s) used	Stateless or Stateful? Why?
Org list or grid	ListView.builder, GridView.builder, LayoutBuilder	Stateful. The list changes while visible, and the screen causes the change through search, swipe, favorite, and reload. The data lives in OrgListScreen's State.
Favorite star	Icon inside OrgCard	Stateless. The star changes, but the parent causes it. OrgCard receives isFavorite from OrgListScreen.
Banner image	Image.network inside OrgBanner	Stateless. OrgBanner stores nothing. The org gives it the URL, and Image.network handles loading and error through its builders.
Submit button	FilledButton in JoinFormScreen	Stateful. Whether it works depends on _agreed, which the form's State owns. The checkbox calls setState() and the button rebuilds.